

Jahid Hasan

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SUMMARY

Modern data scientists devote significant effort to data-intensive tasks such as data discovery, cleaning, and pipeline optimization and debugging. My research rethinks this paradigm by developing intelligent, data-centric systems that automate these stages. I focus on robust frameworks for data discovery, preparation, and pipeline debugging, emphasizing fairness, reliability, robustness, and adaptability under deployment-time adversities. My work integrates machine learning, database systems, and agentic AI to build scalable solutions for ML systems.

EDUCATION

Doctor of Philosophy (Ph.D.) — Computer and Information Technology AUG 2023 - MAY 2027
Purdue University, West Lafayette, IN, USA GPA: 3.94
Supervisor: [Dr. Romila Pradhan](#)
Dissertation: *Intelligent Data-Centric Systems for Fair, Robust, and Adaptive Data Science Pipelines*

Master of Science (M.S.) — Mathematics JAN 2012 - JUL 2023
University of Dhaka, Dhaka, Bangladesh GPA: 3.81
M.S. Dissertation Title: *Age Distributed model of Communicable Disease in Heterogeneous Environment*
Supervisor: [Dr. Md. Kamrujjaman](#)

Bachelor of Science (B.S.) — Mathematics JAN 2017 - DEC 2021
University of Dhaka, Dhaka, Bangladesh GPA: 3.69

SKILLS SET

Programming: **Python, MATLAB, SQL, R, Fortran, SAS, SPSS**
Machine Learning: **SKLearn, PyTorch, TensorFlow, HuggingFace, LLMs, NLP**
Information Retrieval: **Agentic system, RAG, LangChain, Embedding, Vector Database**
Data & Systems: **Pandas, NumPy, MySQL, MongoDB**
Visualization: **Matplotlib, Seaborn, Tableau**

WORK EXPERIENCE

Responsible Data Science Lab, Purdue University — Graduate Research Assistant AUG 2023 - PRESENT
Leading research on reliable and responsible AI systems through the [NSF Award 2237149](#) and [CASMI](#)-funded project, *Data Preparation for Fair and Trustworthy Machine Learning*. Collaborating with researchers from AT&T Research, Cornell University, Université Paris Cité, and the Purdue Virtual Reality Lab.

Selected Research Projects:

Agentic AI for Data Science Pipeline Debugging Ongoing

- Developing a multi-agent framework for diagnosing and repairing data science pipelines affected by data-quality problems, component failures, and suboptimal configurations.
- Coordinating stage-level agent decisions so that local diagnoses and repairs collectively improve global pipeline reliability and performance.

PIPELINS: Identifying Interventions for Resolving Malfunctioning Data Science Pipelines [1] VLDB 2026

- Developed a causal debugging framework for identifying targeted interventions in malfunctioning database, machine learning, and retrieval-augmented generation pipelines.
- Leveraged data profiles and historical executions to recommend interpretable repairs, achieving **5–40%** improvement over LEARN2CLEAN and BUGDOC.

DATASIFT: Selective Data Expansion for Model Performance [2] EDBT 2026

- Designed a reinforcement learning-based framework that identifies and acquires high-value training samples through adaptive data valuation.
- Improved data-acquisition efficiency, predictive performance, and fairness by **30–60%** compared with AUTODATA, SLICETUNER, random sampling, and entropy-based selection.

SENTINEL: Evaluating Pipeline Robustness Under Distributional Shifts [3] IEEE ICDE 2026

- Developed component- and pipeline-level robustness measures for explaining how optimized machine learning pipelines degrade under deployment-time distribution shifts.
- Enabled targeted pipeline reconfiguration and repair without requiring complete model retraining.

FANSHIELD: Predictive Obstacle Assessment for Outdoor Extended Reality Submitted @ ACM CHI 2027

- Developed a LSTM-based trajectory prediction system using VR headset, gaze, controller, and eyeglass motion signals.
- Designed anticipatory safety mechanisms that identify future user movement and reduce collision risks in immersive environments.

Computer and Information Technology, Purdue University — Teaching Assistant SPRING 2026 -

CNIT 39200: Enterprise Data Management

- Instructed four database laboratory sections as an alternate teaching assistant, supporting **76 students** throughout the semester.

Age Distributed model of Hepatitis-B disease in Heterogeneous Environment

- Developed and analyzed an age-structured communicable disease model under heterogeneous environmental and population conditions.
- Conducted modeling, dynamical systems analysis, and data analysis for a project funded by the Bangladesh Ministry of Science and Technology.

RELEVANT PROJECTS

Conversational Shopping Assistant <https://github.com/jahid674/In-sentence-pun-detection>

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline to interpret user intent and generate context-aware product recommendations
- Designed a latency-minimization framework integrating LLMs with vector retrieval to ensure efficient and scalable real-time responses

NLP: In-Sentence Pun Detection <https://github.com/jahid674/In-sentence-pun-detection>

- Developed a system to detect in-sentence puns using contextual embeddings and linguistic features to capture semantic ambiguity
- Applied transformer-based representations to distinguish multiple word senses and evaluated performance using standard NLP metrics

Machine Learning Bias Mitigation Through Data Transformation <https://github.com/jahid674/ML-Bias-Mitigation>

- Implemented data transformation techniques to enforce fairness constraints such as equal opportunity and reduce bias in predictive models
- Conducted comparative analysis across datasets and models to evaluate trade-offs between fairness and predictive performance

LEADERSHIP INVOLVEMENT

- **Senator**, Purdue University (2026–Present). Elected to representing my department in the Purdue Graduate Student Government–representing body of over 12000 graduate students, voicing student concerns, facilitating communication, and contributing to legislative and policy decisions affecting graduate students.
- **Event Director**, Polytechnic Graduate Student Organization (2024–2025). Co-founded this organization for 1,100+ techie graduate students at Purdue by drafting bylaws and building its structure, while leading events, budgets, and logistics that made it popular among students.
- **Fundraising Chair**, Bangladesh Student Association at Purdue (2024–2025). Led fundraising strategy that secured \$56,000 from Purdue’s Student Fee Advisory Board (SFAB) to support cultural exchange activities, while organizing events and strengthening donor relationships to advance the organization’s mission.

AWARDS & PROFESSIONAL ACTIVITIES

- **Participant**, International Conference on Extended Database Technology 2026, **EDBT**, Tampere, Finland, 2026.
- Awarded **US Microsoft EDBT Fellowship**, USA, 2025.
- **Participant**, International Conference on Management of Data 2025, **ACM SIGMOD PODS 2025**, Berlin, Germany, 2025.
- **Attendee**, EDBT Summer School on AI and Data Management, Nicosia, Cyprus, 2025
- Awarded **Travel grant** for the workshop on **Digital Safety**, The Center for Advancing Safety of Machine Intelligence, UL Research Institute. May 2024.
- Awarded **National Science and Technology Fellowship (NST)**, Bangladesh, 2022. A fellowship by the Science and Technology Ministry of Bangladesh.
- **Merit Scholarship**, University of Dhaka, Bangladesh, 2022. A scholarship for 95th percentile academic results.

LICENSES & CERTIFICATIONS

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| • EDBT–Microsoft Summer School on AI and Data Managemen , EDBT Association, Nicosia, Cyprus | July 2025 |
| • Responsible Conduct of Research (RCR) Training , CITI Program | Feb 2024 – Feb 2028 |
| • Social & Behavioral Research Investigators Certification , CITI Program | Feb 2024 – Feb 2028 |
| • Advanced Learning Algorithms , DeepLearning.AI, Stanford University | 2024 |
| • Unsupervised Learning, Recommenders, Reinforcement Learning , DeepLearning.AI, Stanford University | 2024 |
| • Supervised Machine Learning: Regression and Classification , DeepLearning.AI, Stanford University | 2023 |
| • Machine Learning with Python: Foundations , LinkedIn Learning | 2023 |

PUBLICATIONS

[1] Jahid Hasan, Stanley Jiang, Tejendra Singh, Sainyam Galhotra, Romila Pradhan, and Divesh Srivastava. PipeLens: Identifying interventions for resolving malfunctioning data science pipelines. *Proceedings of the VLDB Endowment*, 19, 2026. To appear; presented at the 52nd International Conference on Very Large Data Bases (VLDB 2026).

[2] Jahid Hasan and Romila Pradhan. Selective data expansion for model performance. In *Proceedings of the International Conference on Extending Database Technology (EDBT)*, Tampere, Finland, 2026. OpenProceedings.

[3] Jahid Hasan, Jingya Wang, and Romila Pradhan. SENTINEL: Evaluating Pipeline Robustness to Distributional Shifts. In *Proceedings of the 42nd IEEE International Conference on Data Engineering (ICDE)*, Montreal, Canada, 2026. Accepted to appear.

REFERENCES

- **Dr. Romila Pradhan**
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- **Dr. Divesh Srivastava**
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- **Dr. John Springer**
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